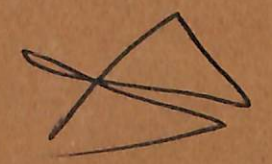


つゆ物

伊勢 喜



A4サイズ 無地 40枚

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①. 002 式 2 (1) 以 定 理

Dirichlet kernel

$$a + ar + ar^2 + \dots + ar^n = \frac{a(1-r^{n+1})}{1-r}$$

$$\begin{aligned} D_k(x) &= \sum_{m=-k}^k e^{imx} \\ &= \frac{e^{i(k+\frac{1}{2})x} - e^{i(-k+\frac{1}{2})x}}{e^{i(\frac{1}{2}x)} - e^{i(-\frac{1}{2}x)}} \\ &= \frac{\sin((k+\frac{1}{2})x)}{\sin(\frac{x}{2})} \end{aligned}$$

$$\sum_{n=0}^n D_n(x) = \frac{1}{2} \left(\sum_{n=0}^n (e^{i(k+\frac{1}{2})x} - e^{i(-k+\frac{1}{2})x}) \right)$$

$$\sum_{n=0}^n e^{i(k+\frac{1}{2})x} = e^{i(\frac{1}{2}x)} \frac{1 - e^{i(n+1)x}}{1 - e^{ix}}$$

$$\sum_{n=0}^n e^{i(-k+\frac{1}{2})x} = \sum e^{-i\frac{1}{2}x} \cdot e^{-i(n)x}$$

$$= e^{-i\frac{1}{2}x} \frac{1 - e^{i(n+1)x}}{1 - e^{-ix}}$$

$$\sum_{n=-\infty}^{\infty} P_n(x)$$

$$= \frac{1}{(z)} e^{i(\frac{1}{2}x)} \frac{1 - e^{i(n+1)x}}{1 - e^{ix}} - e^{-i\frac{1}{2}x} \frac{1 - e^{-i(n+1)x}}{1 - e^{-ix}}$$

$$= \frac{1}{e^{i(-\frac{1}{2}x)}} \frac{1 - e^{i(n+1)x}}{1 - e^{ix}} - \frac{1}{e^{i\frac{1}{2}x}} \frac{1 - e^{-i(n+1)x}}{1 - e^{-ix}}$$

$$= \frac{e^{i(n+1)x} - 1}{e^{i(-\frac{1}{2}x)}} - \frac{1 - e^{-i(n+1)x}}{e^{i\frac{1}{2}x}}$$

$$= \frac{1}{(z)^2} (e^{i(\frac{n+1}{2}x)} + e^{-i(\frac{n+1}{2}x)} - 2)$$

$$= \frac{1}{(z)^2} (e^{i(\frac{n+1}{2}x)} - e^{-i(\frac{n+1}{2}x)})^2$$

$$= \left(\frac{e^{i(\frac{n+1}{2}x)} - e^{-i(\frac{n+1}{2}x)}}{e^{i\frac{1}{2}x} - e^{-i\frac{1}{2}x}} \right)^2$$

$$= \left(\frac{\sinh\left(\left(\frac{n+1}{2}\right)x\right)}{\sinh\left(\frac{x}{2}\right)} \right)^2$$

$$2\pi \mathcal{P}_r(x) = \sum_{-\infty}^{\infty} r^{|n|} e^{inx}$$

$$= \sum_{-\infty}^{-1} r^{|n|} e^{inx} + 1 + \sum_1^{\infty} r^{|n|} e^{inx}$$

$$= \sum_1^{\infty} r^n e^{-inx} + \sum_1^{\infty} r^n e^{inx} + 1$$

$$= \frac{r e^{-ix}}{1 - r e^{-ix}} + \frac{r e^{ix}}{1 - r e^{ix}} + 1$$

$$= \frac{(1 - r e^{-ix})(1 - r e^{ix})}{r e^{-ix}(1 - r e^{ix}) + (1 - r e^{-ix})r e^{ix}} + 1$$

$$= \frac{r e^{-ix} - r^2 + r e^{ix} - r^2}{1 + 1 - r(e^{ix} + e^{-ix}) + r^2} + 1$$

$$= \frac{1 - r^2}{1 - 2 \cos x + r^2}$$

$$D_k(x) = \sum_{-k}^k e^{imx}.$$

$$= \sum_{-k}^{-1} e^{imx} + 1 + \sum_1^k e^{imx}.$$

$$= \sum_{m=1}^k e^{-imx} + \sum_1^k e^{imx} + 1$$

$$= \frac{e^{-ix}(1-e^{-ikx})}{1-e^{-ix}} + \frac{e^{ix}(1-e^{ikx})}{1-e^{ix}} + 1$$

$$= \frac{e^{-i\frac{1}{2}x}(1-e^{-ikx})}{e^{-i\frac{1}{2}x}-e^{-i\frac{3}{2}x}} + \frac{e^{i\frac{1}{2}x}(1-e^{ikx})}{e^{i\frac{1}{2}x}-e^{i\frac{3}{2}x}} + 1.$$

$$= \frac{e^{i(k+\frac{1}{2})x} - e^{-i(k+\frac{1}{2})x}}{e^{i\frac{1}{2}x} - e^{-i\frac{1}{2}x}} + 1$$

$$= \frac{e^{i(k+\frac{1}{2})x} - e^{-i(k+\frac{1}{2})x}}{e^{i\frac{1}{2}x} - e^{-i\frac{1}{2}x}}.$$

$$= \frac{\sin((k+\frac{1}{2})x)}{\sin \frac{x}{2}}.$$

\begin{cases} \text{zero} \end{cases}

P : Polynomial. f is cont.

$$H = \int_k f(x) P(x-\theta) dx$$

is $\mathcal{O}(\frac{1}{x})$.

\end{cases}

$$\int_{-\infty}^{\infty} f(x) W_n(x-\theta) dx.$$

$$\{E_{n,k}\} \rightarrow W(k).$$

$$L = k - k$$

$$\|E_{n,k} - W_n\|_L < \epsilon.$$

$$\left| \int_{-\infty}^{\infty} W_n(x) f(x-\theta) dx - \int_{-\infty}^{\infty} f(x) dx \right|$$

$$\leq \int_{-\infty}^{\infty} |f(x-\theta)| dx \rightarrow 0.$$

L_n .

$$H_1 = \int f(t) L_n(x-t) dt.$$

affinities,

$$A_n = dn \cos^n x.$$

$$H_1 = \int f(t) A(x-t) dt$$

$A(x-t)$ - 解法.

H も 解法

→ ~~$\cos^n x \cos^n x$~~
 $\cos^k x \sin^k x$ 解法.

com.

$F \neq f$ は, 三角関数.

$\sin(n\pi), \cos(n\pi),$
の 和 → 解法.

P_r も 同様.

~~3月~~

距離空間の性質.

① $\downarrow$ の continuity, $\downarrow (A, x)$ の continuity.

② $Z = C_b(X, T)$ の
距離空間
(一般距離空間)

③ $X^\wedge = C_b(X, \mathbb{R})$.
 $X \subset X^\wedge$.
 $X^\wedge \neq X$.
 $C_b(X)$ は X の 完備化.

④ テスト.

1. 2. 3.

$$d(x, a) < \delta$$

$$d(a, b) < \delta$$

$$\Rightarrow |d(x, a) - d(a, b)| < \epsilon$$

$$\delta < \frac{\epsilon}{2}$$

ϵ in
1. 2. 3.

$$|d(x, a) - d(a, b)|$$

$$\leq |d(x, a) - d(x, b) + d(x, b) - d(a, b)|$$

$$\leq |d(x, a) - d(x, b)|$$

$$+ |d(x, b) - d(a, b)|$$

$$\leq d(x, b) + d(x, a) < 2\delta < \epsilon$$

1. 2. 3.

$$|d(x, A) - d(a, A)| \leq d(x, a)$$

1. 2. 3.

2.

$$\mathcal{C}_b(X, Y)$$

top.

metrical.

$$f_n \rightarrow f$$

$$|f(x) - f(a)|$$

$$\leq |f(x) - f_n(x)|$$

$$+ |f_n(x) - f_n(a)|$$

$$+ |f_n(a) - f(a)|$$

$$< 3 \cdot \frac{\epsilon}{3} = \epsilon \quad \square$$

$$f_n \rightarrow f \Rightarrow f \text{ is continuous}$$

$\mathbb{R}$ is complete.

$$d(f_n, f)$$

$$D(f_n, f_m) < \epsilon.$$

$$f(x), f_n(x), \dots$$

$$f_n(x) \rightarrow f(x),$$

$$f_n.$$

$$d(f_n(x), f_m(x)) \leq D(f_n, f_m) < \epsilon$$

$\downarrow$ Cauchy's.

$$d(f_n(x), f(x)) \leq D(f_n, f) < \epsilon.$$

$$D(f_n, f) < \epsilon.$$

完備性の定義 Σ と f_n 列

とある。 $\lim$ $\rightarrow$ $\lim$

der seq

$$A \subset \mathbb{R}$$

closed.

A is complete.

$$A \subset \mathbb{R}$$

compact.

A is closed

interval.

$M \geq 2$.

$$f = \sum_{n=0}^{\infty} f_n(x).$$

f_n is continuous.

$$\sup |f_n| \leq M_n.$$

$$\sum M_n < \infty.$$

$$\sum f_n \Rightarrow f. \text{ by } \frac{1}{M_n}$$

Stone - Weierstraß の
定理

$X \subset \mathbb{C}^n$

$A \subset \mathbb{C}^n$

- ① $1 \in A$ ② $x \neq y \Rightarrow \exists f \in A, f(x) \neq f(y)$
 ③ $f, g \in A \Rightarrow f+g \in A$

$$\Rightarrow \mathcal{C}(A) = \overline{A}$$

(- $\mathbb{R}$, μ)
 $F_{a,b}(a+b) \in \mathbb{R}, F_{a,b} \in A$

$$F_{a,b}(a) = f(a)$$

$$F_{a,b}(b) = f(b)$$

...

$$f(a) \neq f(b), \text{ s.t. } x$$

$$F_{a,b}(x) = (f(b)-f(a)) \frac{g(x)-g(a)}{g(b)-g(a)} + f(a)$$

$$\boxed{F_{a,a} = f(a)} \quad F_{a,b} \in A, (A \in A \text{ s.t. } x, y, z)$$

$$V_{a,b} = \{x \mid F_{a,b}(x) < f(x) + \varepsilon\}$$

$$a \in V_{a,b}$$

$$\mathbb{R} = \bigcup V_{a,b}$$

$$\mathbb{R} = \bigcup_{n=1}^{\infty} V_{a_n, b_n}$$

$$G_b = \min_i \{F_{a_i, b}\}$$

$$\text{z.z. } G_b(x) \leq F_{a,b}(x) < f(x) + \varepsilon$$

$$G_b(b) = f(b)$$

$$G_b(x) < f(x) + \varepsilon$$

$$V_b = \{x \mid f(x) - \varepsilon < G_b(x)\}$$

$$H = \max_{b_i} G_{b_i}$$

$$\text{z.z. } f(x) - \varepsilon < H(x) \leq G_b(x) < f(x) + \varepsilon$$

$$\|H - f\| < \varepsilon, H \in A$$

Diniの定理

$$f_n \leq f_{n+1},$$

$$f_n \rightarrow f$$

$$\Rightarrow f_n \nearrow f.$$

$$\text{for } n \geq n_0, f_n \leq f_{n+1} \leq f.$$

$$f - f_n \searrow f - f_{n+1} \searrow \dots$$

$$\forall x, \exists n, |f_n(x) - f(x)| < \epsilon,$$

$$\downarrow$$

$$\bigcup_{n \geq n_0} \exists x,$$

...

$$\bigcup_{n \geq n_0} \bigcup_{k \geq n} \dots$$

$$\bigcup_{k_1, x_1} \dots \bigcup_{k_n, x_n}.$$

$$k \geq \max k_i, k < n.$$

$$x \in \bigcup_{k_i} x_i, f_{k_i} \leq f_k.$$

$$\epsilon > f(x) - f_{k_i}(x) \geq f(x) - f_k(x) \\ \geq f(x) - f_k(x).$$

$$k < n \Rightarrow \|f - f_n\| < \epsilon.$$

$$[0, 1] \text{上 } \sqrt{x} \text{ の } \epsilon \text{ 近傍}$$

$$P(x) \sim P(0) = 0.$$

(wiegen $\epsilon \rightarrow 0$).

$$\|P - \sqrt{\cdot}\| < \frac{\epsilon}{2}.$$

$\mathcal{A}^n$ の ϵ 近傍.

$$P(0) = 0, \sqrt{0} = 0.$$

$$|x| = |P(0) - 0| \leq \frac{\epsilon}{2}$$

$$Q = P - \sqrt{\cdot}.$$

$$\|Q - \sqrt{\cdot}\| \leq \|P - \sqrt{\cdot}\| + |x| < \epsilon.$$

$$z_0 = 0.$$

$$P_{n+1}(t) = P_n(t) + \frac{1}{2}(t - P_n^2(t)),$$

$$0 \leq P_n \leq \sqrt{t}.$$

$$\sqrt{t} - P_{n+1}$$

$$= \sqrt{t} - P_n(t) - \frac{1}{2}(t - P_n^2)$$

$$= \sqrt{t} - P_n - \frac{1}{2}(\sqrt{t} - P_n)(\sqrt{t} + P_n),$$

$$= (\sqrt{t} - P_n) \left(1 - \frac{1}{2}(\sqrt{t} + P_n)\right),$$

$$P_n \leq \sqrt{t} \quad - \quad I_n \geq -\sqrt{t}$$

$$\geq (\sqrt{t} - P_n) \left(1 - \frac{1}{2}2\sqrt{t}\right),$$

$$\geq (\sqrt{t} - P_n)(1 - \sqrt{t}) \geq 0$$

$$\forall n, \quad 0 \leq P_n \leq \sqrt{t}.$$

$$P_{n+1} - P_n = \frac{1}{2}(t - P_n^2) \geq 0.$$

$$P_n \leq P_{n+1}, \quad P_n(t) \leq \sqrt{t}$$

$$P = P + \frac{1}{2}(t - P^2)$$

$$t = P^2 \quad \Rightarrow \quad P = \sqrt{t}$$

$$P \Rightarrow \sqrt{t}$$

Spencer widerstellt.

$$(1) \quad 1 \in A.$$

$$(2) \quad \forall x \neq 0. \quad \exists f \in A \quad f(x) \neq f(0).$$

$$\Rightarrow \bar{A} = C(x)$$

$\bar{A}$ ist, $\mathbb{R}$ ist $\mathbb{R}$, $\mathbb{R} \in \mathbb{R}$

$$f, g \in \bar{A} \Rightarrow f \vee g \in \bar{A}$$

$$\max(f, g) = \frac{f+g + |f-g|}{2}.$$

$$|f| \geq \sqrt{f^2}.$$

$$P_n\left(\frac{f^2}{\|f\|^2}\right) \rightarrow \frac{\sqrt{f^2}}{\|f\|}.$$

how $\mathbb{R}$ is $\mathbb{R}$.

$$\bar{A} = \overline{\bar{A}} = C(x).$$

L.H.C.S.P. の Stone-Weierstraß

$$C_0(X) = \{f \mid f(\infty) = 0\}.$$

$$A \text{ かつ } A \subset C_0(X)$$

$$1. \forall x. \exists f \in A. f(x) \neq 0.$$

$$2. \forall x, y. \exists f \in A. f(x) \neq f(y).$$

$$\overline{A} \subset C_0(X).$$

$$B = \langle A, 1 \rangle.$$

$$f \in C(X). \exists g: \text{明 } \forall n.$$

$$\|f - g(f_1, f_2, \dots, f_n)\| < \epsilon.$$

$$G(X) = \{g\} \text{ (} \mathcal{G} \text{)}$$

$$G(\infty) = g(0, \dots, 0).$$

$$|x| < \frac{\epsilon}{2}. \quad \|f - H\| < \epsilon$$

これは $\mathcal{A}$ の性質である。

$$\textcircled{1} - 1 \in A.$$

$$\textcircled{2} - \forall x. \exists f. f(x) \neq f(0).$$

$$\textcircled{3} f \in A \Rightarrow \overline{f} \in A.$$

$$\|f\| \leq \| \operatorname{Re}(f) \| + \| \operatorname{Im}(f) \|$$

$$\|f\| \leq \| \operatorname{Re}(f) \| + \| \operatorname{Im}(f) \|.$$

Ex. 4.1.3

測度算術,

$$\mathbb{R} = [-\infty, +\infty],$$

上の演算.

$$+\infty \pm a = +\infty \quad a \leq +\infty$$

$$-\infty \pm a = -\infty \quad -\infty \leq a$$

$$a + \infty = \infty, a = \begin{cases} \infty \\ 0 \\ -\infty \end{cases} \quad \frac{1}{\infty} = 0.$$

$$\frac{1}{0} = 0, \quad 0 \cdot \infty = 0, \quad \infty \cdot (-\infty) = -\infty$$

$$\infty - \infty \neq 0, \quad \frac{\infty}{\infty} \neq \frac{1}{2} \neq \frac{1}{2} \neq \frac{1}{2}$$

$$(-1)(+\infty) = -\infty$$

$$(-1)(-\infty) = +\infty$$

$$B = (B \setminus A) \sqcup A.$$

$$\mu(B) \leq \mu(B \setminus A) + \mu(A) \\ \leq \mu(A).$$

$$A \subset B.$$

$$\mu(A) \leq \mu(B).$$

$$\mu(B) < \infty, \quad \mu(A) < \infty.$$

$$\Rightarrow \mu(B - A) = \mu(B) - \mu(A).$$

$$A \subset \bigcup_{n=1}^{\infty} B_n.$$

$$\Rightarrow \mu(A) \leq \sum_{n=1}^{\infty} \mu(B_n).$$

$$\bigcup_{n=1}^{\infty} B_n = B_1 \sqcup \bigsqcup_{n=2}^{\infty} (B_n \setminus B_{n-1}).$$

$$A_n \subset A_{n+1}, \quad A = \bigcup_{n=1}^{\infty} A_n$$

$$\mu(A) = \lim_{n \rightarrow \infty} \mu(A_n)$$

$$B_{n+1} \subset B_n, \quad \mu(B_1) < \infty \\ \mu(B_1) = \lim_{n \rightarrow \infty} \mu(B_n)$$

$$n \geq 1 \quad A \cup B.$$

~~over~~

$$= (A \setminus B) \cup B.$$

$$\mu(A \cup B) \leq \mu(A \setminus B) + \mu(B)$$

Recursion $\leq \mu(A) + \mu(B)$

$\forall n,$

$$\mu\left(\bigcup_{k=1}^n B_k\right) \leq \sum_{k=1}^n \mu(B_k)$$

$n \rightarrow \infty$

$$\mu(A) \leq \mu\left(\bigcup_{k=1}^{\infty} B_k\right) \leq \sum_{k=1}^{\infty} \mu(B_k).$$

$$B_n = B_q = A_q.$$

$$B_n = B_{n+1} \setminus B_n.$$

$$\bigcup A_n = \bigsqcup B_n. \quad \mu(A_n) < \infty$$

$$\tilde{\bigcup} A_n = \bigsqcup_{k=1}^{\infty} B_k.$$

$$\mu(\tilde{\bigcup} A_n)$$

$$= \sum \mu(B_k)$$

$$= \mu(B_n).$$

$$\mu(\tilde{\bigcup}) = \sum_{k=1}^{\infty} \sum_{i=1}^n \mu(B_k) \\ = \lim_{n \rightarrow \infty} \sum_{k=1}^n \mu(B_k)$$

第 1 部 完

第2部.
司則の証明

①. $f: (X, \mu) \rightarrow (\mathbb{R}, \mathcal{O})$
 $\mathcal{O} = \sim$ 有界な関数
 同値関係

of. $f+g, f-g, f \cdot g$
 $\frac{1}{g}, h(f, g)$

sup inf -
 lim sup, lim inf -
 lim

手書き

第3部, 3.2.1. ~~2.4.1~~ $\sup f$

$$\int s + t \, d\mu = \int s \, d\mu + \int t \, d\mu$$

$$\int \alpha s \, d\mu = \alpha \int s \, d\mu$$

Fatouの補題

$$\int \liminf f_n \, d\mu \leq \liminf \int f_n \, d\mu$$

$$\liminf_{n \rightarrow \infty} a_n = \sup_n \left(\inf_{k \geq n} a_k \right)$$

$$= \lim_{n \rightarrow \infty} \left(\inf_{k \geq n} a_k \right)$$

$$g_n(x) = \inf_{k \geq n} (f_k(x))$$

$$\lim g_n(x) = \liminf f_n(x)$$

$$\int \liminf f_n \, d\mu = \liminf \int f_n \, d\mu$$

$$222, \quad g_n = \inf_{k \geq n} f_k(x)$$

$$k \geq n \Rightarrow g_n \leq f_k(x).$$

$$k \geq n \Rightarrow \int_{\mathcal{D}} g_n d\mu \leq \int_{\mathcal{D}} f_k d\mu$$

$n \rightarrow \infty$

$$\int_{\mathcal{D}} g_n d\mu \leq \left(\inf_{k \geq n} \int_{\mathcal{D}} f_k d\mu \right)$$

$$\int_{\mathcal{D}} \liminf_{n \rightarrow \infty} f_n d\mu \leq \liminf_{n \rightarrow \infty} \int_{\mathcal{D}} f_n d\mu$$

D

$$\alpha < \int_{\mathcal{D}} s.$$

$$\beta < \int_{\mathcal{D}} t.$$

$$\int_{\mathcal{D}} s + t \leq \int_{\mathcal{D}} (s + t) d\mu$$

$$\alpha + \beta < \int_{\mathcal{D}} (s + t) d\mu$$

$$\int_{\mathcal{D}} s + \int_{\mathcal{D}} t \leq \int_{\mathcal{D}} (s + t)$$

$$\begin{array}{l} u \leq f \leq s. \\ \quad \quad \quad s \leq t. \\ \hline u \leq s. \end{array}$$

★ 星型領域は、 $\mathbb{R}^n$ 上で微分可能。

Def. 下連続関数

$$f: (X, d) \rightarrow \mathbb{R} = [-\infty, +\infty)$$

$$\forall x \in X, \forall \varepsilon > 0, \exists \delta > 0, \forall y \in U_\delta(x), f(y) > f(x) - \varepsilon,$$

$$(\Leftrightarrow) f^{-1}([a, +\infty)) \text{ is open, or } f^{-1}([-\infty, a]) \text{ is closed.}$$

Prop. f : 下連続関数, $\forall x, f(x) \geq s > 0$.

$$\exists \{f_n\}_{n=1}^{\infty} \text{ such that } f_n \geq s.$$

$$f_n \uparrow f_{n+1}, \quad f_n \rightarrow f$$

$$f_n(x) = \inf_{p \in X} \{f(p) + n d(p, x)\}$$

条件を列挙

$$(i) \quad \overline{f_n(x) \leq f(x)} \quad (p=x \text{ かつ } 1)$$

$$(ii) \quad \underline{f_n(x) = f}$$

$$f(p) + n d(p, x) \leq f(p) + (n+1) d(p, x) \quad \text{かつ}$$

$$f_n \leq f_{n+1}$$

$$(iii) \quad f(p) + n d(p, x) \geq f(p) \geq s \quad \text{かつ}$$

$$f_n(x) \geq s$$

(iv) f_n の連続性

$$\forall x, \forall \varepsilon > 0, \exists \delta, \forall y \in U_\delta(x), f_n(y) \leq f(p) + n d(p, x)$$

$$f_n(y) - f_n(x) < n(d(p, y) - d(p, x)) - \varepsilon$$

$$\leq n d(x, y) - \varepsilon$$

$$f_n(y) - f_n(x) \leq n d(x, y)$$

$$\therefore |f_n(x) - f_n(y)| \leq n d(x, y)$$

(連続性)

⑤. $f_n \nearrow f$ $\forall x \in \mathbb{R}$. $x: f'_x$.
 $d(p, x) < \delta$. $n \geq N$

$f(p) + n d(p, x)$
 $\left(\begin{array}{l} \geq f(p) > f(x) - \epsilon, \\ \text{inf.} \end{array} \right.$

~~$f_n(x) \geq f(x) - \epsilon$~~

$d(p, x) \geq \delta$. $N \delta > f(x) - \epsilon$,

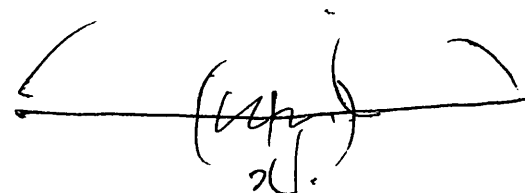
$n \geq N \Rightarrow$

$f(p) + n d(p, x) \geq n d(p, x) > \delta \cdot N$
 $> f(x) - \epsilon$

$\Rightarrow f_n(x) > f(x) - \epsilon$

$f(x) \geq f_n(x) > f(x) - \epsilon$

$f(p) + n d(p, x) > N \delta > f(x) - \epsilon$
 $\therefore f_n(x) \rightarrow f(x)$
 $n \rightarrow \infty$



① $f(p) + n d(p, x) > f(x) - \epsilon$

Fact.

M is a compact manifold.

$f: M \rightarrow \mathbb{R}$ is continuous.

$f \in C^\infty$. $C^\infty \subset C^0$.

$V_\epsilon = \{x \mid |f(x) - f(a)| < \epsilon\}$

$a \in V_\epsilon$. $M = \bigcup_a V_\epsilon$

$\therefore M = V_{\epsilon_1} \cup \dots \cup V_{\epsilon_n}$

$\because$ $S = \{x_i\} \subset V_{\epsilon_i}$ $\forall i$.

λ_i - partition of unity
 $\sum \lambda_i = 1$

$g(x) = \sum_{i=1}^n f(x_i) \lambda_i$. $x \in V_i$

$|f(x) - g(x)| = \left| \sum_{i=1}^n \lambda_i f(x_i) - g(x) \right|$

$\leq \sum_{i=1}^n \lambda_i |f(x_i) - f(x)|$

$< \epsilon \cdot \sum \lambda_i \leq \epsilon$

$\|f - g\|_\infty \leq \epsilon$

$f: M \rightarrow \mathbb{R}$, f 下連続.

$$f \geq s > 0,$$

2nd.

$$\exists \{f_n\}_{n=N}^{\infty} \quad f_n(x) \nearrow f.$$

$$f_n > \frac{s}{2}$$

$$\|f_{n+1} - f_n\| > \frac{1}{2n(n+1)}.$$

$$\frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+1} \right).$$

lem 1 p4. $g_n \nearrow f$. $g_n \geq s$ 連続.

$$G_n := g_n - \frac{1}{n}. \quad \text{2nd. } N \in \mathbb{N}.$$

$$N > \frac{4}{s} \quad \text{2nd. } n \geq N \Rightarrow$$

$$G_n = g_n - \frac{1}{n} \geq s - \frac{1}{N} > s - \frac{s}{4} = \frac{3}{4}s.$$

$$G_n > \frac{3}{4}s.$$

$$G_{n+1} - G_n$$

$$= g_{n+1} - g_n - \frac{1}{n+1} + \frac{1}{n} \geq -\frac{1}{n+1} + \frac{1}{n} = \frac{1}{n(n+1)}$$

$$\therefore G_{n+1} - G_n \geq \frac{1}{n(n+1)}$$

lem 2 p4. $\exists f_n: C^\infty$.

$$\|f_n - g_n\| < \frac{1}{4n(n+1)}$$

$$\|f_n - G_n\| < \frac{1}{4n(n+1)}$$

$$f_n \rightarrow f \text{ if } \phi \neq 0. \quad \frac{1}{(n+1)(n+2)} \leq \frac{1}{n(n+1)}$$

$$f_{n+1} - f_n \geq \underbrace{G_{n+1}}_{-G_n} - \frac{1}{4(n+1)(n+2)} + \frac{1}{4n(n+1)}$$

$$\geq \frac{1}{n(n+1)} - \frac{1}{4(n+1)(n+2)} - \frac{1}{4n(n+1)}$$

$$\geq \frac{1}{n(n+1)} - 2 \cdot \frac{1}{4n(n+1)} = \frac{1}{2n(n+1)}$$

$$\therefore f_{n+1} - f_n \geq \frac{1}{2n(n+1)}$$

$$\frac{1}{4n(n+1)} \leq \frac{1}{n}.$$

$$f_n > g_n - \frac{1}{4n(n+1)} > \frac{3}{4}s - \frac{1}{n} > \frac{3}{4}s - \frac{1}{4}s = \frac{s}{2} \quad (f_n > \frac{s}{2})$$

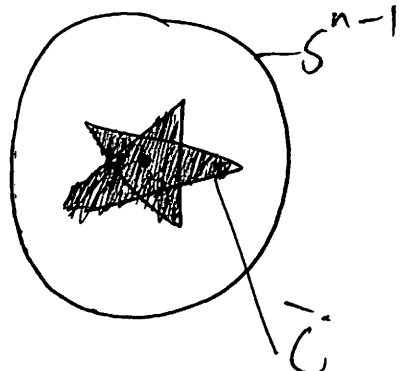
$\left\{ \begin{array}{l} \mathbb{R}^n \stackrel{\text{diff}}{\simeq} B_n \text{ 定理 1.} \\ \text{星型領域 } C \text{ 上有界連続.} \\ \text{証明 30. } C \text{ の中心 } o. \end{array} \right.$

$$C \subset B_n(o).$$

$$S^{n-1} = \partial B_n(o) \text{ 上.}$$

$$f: S^{n-1} \rightarrow \mathbb{R} \text{ 上.}$$

$$x \mapsto \sup_{z \in C} \{ \lambda z \mid \lambda x \in C \}.$$

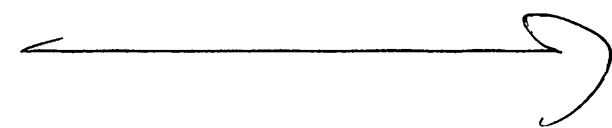


① f は、下半連続. 2.

②. $f \geq S > 0$. (0 $\leq S \leq 1$ 上連続.)

③ は f のかわり (1: δ_1 上連続 $\bar{B}_{\delta_1}(o) \subset C$ となる.)

f の下半連続性を示す



$$\forall \epsilon > 0, \forall x \in S^{n-1},$$

$$\exists \lambda > 0, \lambda x \in C \text{ 上, } \lambda > f(x) - \epsilon.$$

$$\lambda x \in C \text{ 上, } \exists \delta.$$

$$V_\delta(\lambda x) \subset C.$$

$$\cancel{V_\delta = \bigcup_{\lambda} \frac{\delta}{\lambda} (x) \cap S^{n-1} \text{ 上.}}$$

$$V_\delta = \bigcup \left(x; \frac{\delta}{\lambda} \right) \cap S^{n-1} \text{ 上.}$$

V は S^{n-1} の open set 上,

$$z \in V \text{ かつ } S^{n-1} \text{ 上.}$$

$$\|x - z\| < \frac{\delta}{\lambda}, \text{ 上.}$$

$$\|\lambda x - \lambda z\| = \lambda \|x - z\| < \delta, \text{ 上.}$$

$$\lambda z \in C, \therefore f(z) \geq \lambda.$$

$$\forall z \in V, f(z) \geq f(x) - \epsilon.$$

よって f は、

下半連続

f は半径の関数である。

$$(iii) \\ \mathcal{M} = \left\{ (a, b, c, d) \in \mathbb{R}^4 \mid a < b, \quad c < d, \quad d - c > \frac{b - a}{2} \right\}$$

$$h: \mathbb{R} \rightarrow \mathbb{R}^c$$

$$h: \mathbb{R} \times \mathcal{M} \rightarrow \mathbb{R}, \quad C^\infty$$

$$h(x, a, b, c, d), \quad \text{to } \mathbb{R}$$

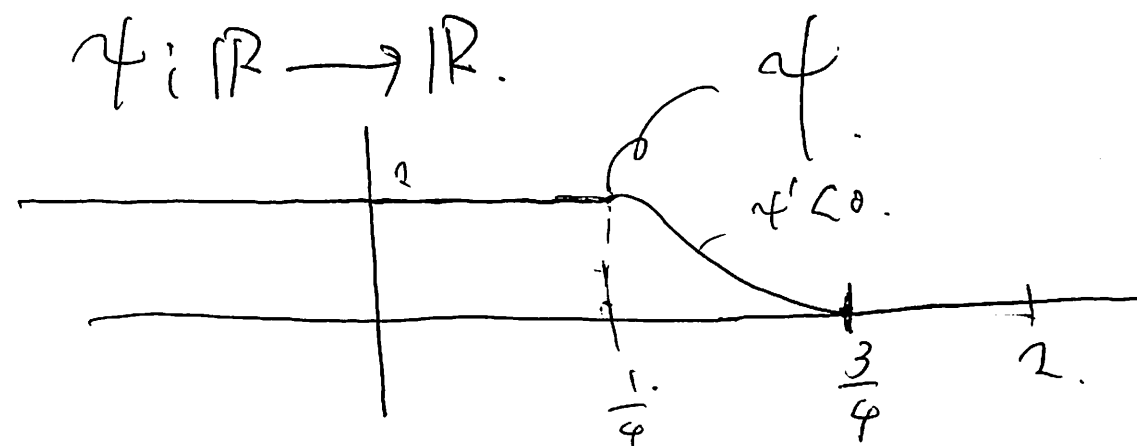
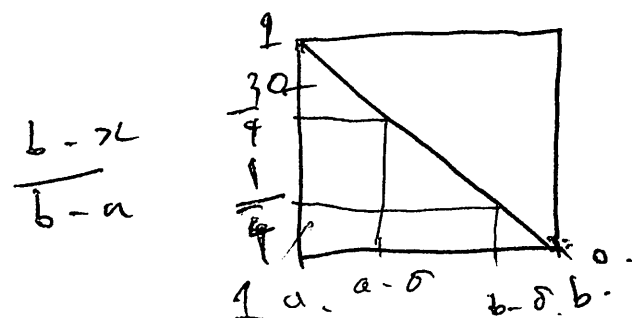
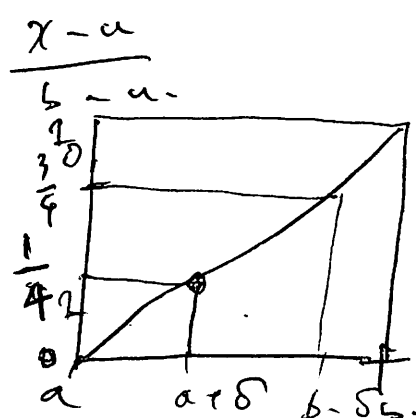
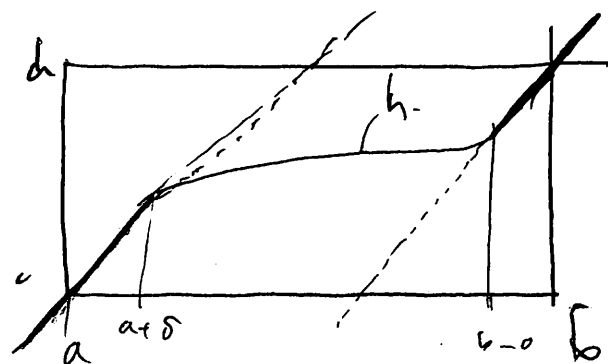
$$\textcircled{1} \partial_x h > 0.$$

$$\textcircled{2} h([a, b]) = [c, d].$$

$$\textcircled{3} \delta = \frac{b-a}{4}.$$

$$x < a + \delta \Rightarrow h(x) = \frac{1}{2}(x-a) + c.$$

$$b - \delta < x \Rightarrow h(x) = \frac{1}{2}(x-b) + d.$$



$$h(x, a, b, c, d) =$$

$$\left(\frac{1}{2}(x-a) + c \right) \phi\left(\frac{x-a}{b-a} \right) + \left(\frac{1}{2}(x-b) + d \right) \left(1 - \phi\left(\frac{x-a}{b-a} \right) \right)$$

$$\partial_x h > 0 \text{ for } x \in [a, b].$$

$$b - \delta < x - a$$

$$b - a - \delta = b - a - \frac{b-a}{4} = \frac{3}{4}(b-a).$$

$$b - \delta < x \Rightarrow \frac{3}{4} < \frac{x-a}{b-a}.$$

$$x - a \leq \delta, \quad \frac{x-a}{b-a} \leq \frac{1}{4}.$$

$$\Rightarrow b - a - \delta < x - a.$$

$$x > a + \delta.$$

$$b - x > b - a - \delta.$$

$$\frac{b-x}{b-a} > 1 - \frac{1}{4} = \frac{3}{4}.$$

$$\left(\frac{1}{2}(x-a) + c \right) \phi\left(\frac{x-a}{b-a} \right) + \left(\frac{1}{2}(x-b) + d \right) \left(1 - \phi\left(\frac{x-a}{b-a} \right) \right)$$

$$\mu > \frac{1}{1-\delta}, \quad \mu \in \mathbb{N}.$$

(Lemma 3 による) $\{f_n\}_{n=\mu}^{\infty} \in \mathcal{B}_1$.

$$\mathcal{U}(0, 1 - \frac{1}{\mu}) \xrightarrow{\sim} B(0, f_{\mu}(x))$$

$$\Phi = \left\{ \frac{x}{\|x\|} h\left(\|x\|, 1 - \frac{1}{\mu}, 1 - \frac{1}{\mu+1}, f_{\mu}\left(\frac{x}{\|x\|}\right), f_{\mu+1}\left(\frac{x}{\|x\|}\right)\right) \right. \\ \left. \frac{x}{\|x\|} h\left(\|x\|, 0, 1 - \frac{1}{\mu}, 0, \frac{s}{2}\right) \right\}$$

$$f_{\mu-1} = \frac{s}{2}$$

$$\Phi(x) = \begin{cases} \frac{x}{\|x\|} h\left(\|x\|, 1 - \frac{1}{\mu}, 1 - \frac{1}{\mu+1}, f_{\mu}\left(\frac{x}{\|x\|}\right), f_{\mu+1}\left(\frac{x}{\|x\|}\right)\right) & (n > \mu) \\ \frac{x}{\|x\|} h\left(\|x\|, 0, 1 - \frac{1}{\mu}, 0, \frac{s}{2}\right) & (n = \mu) \end{cases}$$

かつ $\mathcal{U}(\mu) \cap \mathcal{U}(\mu+1) \neq \emptyset$. $\mathcal{U}(\mu) \neq \emptyset$ より $\mu > 0$.

非線形性を μ^{-3} とする.

$$\mathcal{U}_2 = \begin{pmatrix} \partial_x h & * \\ 0 & 1 \\ \vdots & \vdots \\ 0 & 0 \end{pmatrix}$$

すなわち $\partial_x h \approx \mu^{-3}$. $\mathcal{U}_2$ が μ^{-3} である.

($\mathcal{U}$ は μ の関数であり、角を μ によって変える).

Q. E. D.

ref

Drink Feras, Analysis III,

<http://page.math.fu-berlin.de/feras/AMA/Ana3.pdf>

2015/10月 陶寛

逆函数定理. 隐函数定理.

隐函数定理.

$$\Omega \subset V_1 \times V_2.$$

$$F: \Omega \rightarrow W.$$

Ω 上 C^1 级.

$$F(x_0, y_0) = 0.$$

$$D_2 F(x_0, y_0): V_2 \rightarrow W.$$

如可逆 $\mathbb{R}$. 逆映射.

$$\text{即 } x \in \Omega_1 \in \mathcal{N}(x_0) \\ \Omega_2 \in \mathcal{N}(y_0).$$

$$g: \Omega_1 \rightarrow \Omega_2.$$

$$\text{即 } F(x, g(x)) = 0.$$

$$Dg(x) = - (D_2 F(x, g(x)))^{-1} D_1 (F(x, g(x)))$$

$$F(x, \vartheta) = 0.$$

⟹

$$\begin{aligned} \mathcal{J} &= \mathcal{J} - L_0^{-1} F(x, \vartheta) = G(x, \vartheta) \\ &= \Phi_x(\vartheta). \end{aligned}$$

$$\cancel{\Phi_x(\vartheta(x)) = \Phi(\vartheta)}$$

$$\Phi_x(\vartheta(x)) = \mathcal{J}(x)$$

也 对 $\vartheta \sim \vartheta_0$ 也.

$$\Phi_x(\vartheta) = \mathcal{J} - L_0^{-1} (F(x, \vartheta))$$

故

$$\Phi_x(\vartheta_1) - \Phi_x(\vartheta_2)$$

$$= \mathcal{J}_1 - L_0^{-1} (F(x, \vartheta_1)) - (\mathcal{J}_2 - L_0^{-1} (F(x, \vartheta_2)))$$

$$= L_0^{-1} (D_2 F(x_0, \vartheta_0)(\vartheta_1 - \vartheta_2) - (F(x, \vartheta_1) - F(x, \vartheta_2)))$$

$$\cancel{\|L_0^{-1} (D_2 F(x_0, \vartheta_0)(\vartheta_1 - \vartheta_2))\|}$$

可证.

$$F_2(x_0, \vartheta_0)(\vartheta_1 - \vartheta_2) - (F(x, \vartheta_1) - F(x, \vartheta_2)).$$

$$= F_2(x_0, \vartheta_0)$$

$$\|x - x_0\| \leq \delta_1, \quad \|\vartheta_1 - \vartheta_0\| \leq \eta, \quad \|\vartheta_2 - \vartheta_0\| \leq \eta.$$

$$\|\Phi_x(\vartheta_1) - \Phi_x(\vartheta_2)\| \leq \frac{1}{2} \|\vartheta_1 - \vartheta_2\|$$

$$2\delta_2, \quad \|G\| \quad \|\Phi_x(\vartheta_0) - \Phi_x(\vartheta_0)\| < \frac{1}{2}$$

$$\|x - x_0\| \leq \delta_2, \quad \|G(x, \vartheta_0) - G(x_0, \vartheta_0)\| < \frac{1}{2}$$

$$\|\mathcal{J} - \mathcal{J}_0\| \leq 1$$

$$G(x_0, \vartheta_0) = \mathcal{J}_0$$

$$\|x - x_0\| < \delta = \min \{\delta_1, \delta_2\}$$

$$\|\Phi_x(\vartheta) - \mathcal{J}_0\| = \|G(x, \vartheta) - G(x_0, \vartheta_0)\|$$

$$\leq \|G(x, \vartheta) - G(x, \vartheta_0)\| + \|G(x, \vartheta_0) - G(x_0, \vartheta_0)\|$$

$$\leq \frac{1}{2} \|\vartheta - \vartheta_0\| + \frac{1}{2} \leq 1 \quad \text{也 证.}$$

$$\Phi_x \mathcal{J} = C(B_\delta(x), \bar{B}_1(\vartheta_0))$$

$x \in B_\delta(x)$.

$$\Phi_x: \mathcal{P} \rightarrow \mathcal{J}, \quad \Phi_x(\mathcal{J}(x)) = \mathcal{J}(x)$$

$\mathcal{J}$: 完备.

全微分可能性の証明

$$\lim_{x \rightarrow x_0} \frac{\|f(x) - f(x_0) - L(x - x_0)\|}{\|x - x_0\|} = 0.$$

$$Df = L \quad L = (Df)(x_0).$$

$$f(x) = f(x_0) + L(x - x_0) + \xi(x),$$

$$\lim_{x \rightarrow x_0} \frac{\xi(x)}{\|x - x_0\|} = 0.$$

$$G(x, y) = G(x_0, y_0) + L_1(x - x_0) + L_2(y - y_0) + \xi(x, y).$$

$$F(x, y) = F(x_0, y_0) + L_1(x - x_0) + L_2(y - y_0) + \xi(x, y).$$

$$L_2(y - y_0)$$

$$L_2(y_1 - y_2) = (F(x, y_1) - F(x, y_2)).$$

$$F(x, y_1) - F(x, y_2)$$

$$= L_1(x - x_0) + L_2(y_1 - y_2) + \xi(x, y_1) - \xi(x, y_2)$$

$$= L_2(y_1 - y_2)$$

$$F(x, y_1) = F(x, y_0) + L_2^{(x, y_0)}(y_1 - y_0) + \xi(x, y_1).$$

$$F(x, y_2) = F(x, y_0) + L_2^{(x, y_0)}(y_2 - y_0) + \xi(x, y_2).$$

$$F(x, y_1) - F(x, y_2)$$

$$= F(x, y_0) - F(x, y_0)$$

$$+ L_2^{(x, y_0)}(y_1 - y_2) + \xi(x, y_1) - \xi(x, y_2)$$

$$+ L_2(x, y_1)(y_1 - y_2)$$

$$- L_2(x, y_2)(y_2 - y_0)$$

$$- L_2(x, y_0)(y_1 - y_2)$$

$$L_2(x_0, y_0)(y_1 - y_2) - (F(x, y_1) - F(x, y_2)).$$

$$= (L_2(x_0, y_0) - L(x, y_0))(y_1 - y_2) + \xi(x, y_1) - \xi(x, y_2).$$

$$\xi(x, z_1) + \xi(x, z_2).$$

$$\frac{\|z_1 - z_0\|}{\|z_1 - z_0\|} = \frac{\xi(x, z_1)}{\|z_1 - z_0\|}.$$

$$(\xi) \quad |\xi| \leq \mu \|z_1 - z_0\|.$$

$$z_1 \rightarrow z_2.$$

$$F(x, z_1) = F(x, z_0) + L_2(x, z_0)(z_1 - z_0) + \xi(x, z_1).$$

$$\begin{aligned} F(x, z_1) - F(x, z_2) &= L_2(x, z_0)(z_1 - z_2) \\ &\quad + \xi(x, z_1, z_2) \end{aligned}$$

$$L_2(x_0, z_0)(z_1 - z_2) - (F(x, z_1) - F(x, z_2)).$$

$$= (L_2(x_0, z_0) - L_2(x, z_2))^{(z_1 - z_2)} + \xi(x, z_1, z_2).$$

$$\xi(x, z) \neq 0. \quad J. 2,$$

$$\|\xi(x, z_1) - \xi(x, z_0)\|$$

$$E(x, z_1, x_0, z_0) \neq 0.$$

$$E(x, x_2, z_1, z_2) \neq 0.$$

$$(x, z) \rightarrow (x_0, z_0).$$

$$E(x, x_0, z_1, z_0) \neq 0.$$

$$E(x, x_1, z_1, z_2) \sim \frac{d\tau}{d\tau} = \frac{d\tau}{d\tau}.$$

$$\xi = \frac{E(x, x', z, z)}{\|z - z\|}. \quad (z \neq z).$$

$$\|\xi(x, x', z, z) - \xi(x, x_0, z, z_0)\| < \epsilon.$$

$$|\xi(x, x', z, z)| < |\xi(x, x_0, z, z_0)| + \epsilon.$$

$$|\xi(\sim)| < \mu \|z - z\|.$$

2. 2.

這國好定五里山

陰利好走理

這也. 算1子 !!

$$I: \mathbb{C}^t \rightarrow \mathbb{R} \quad \text{ist} \quad I: \mathbb{C} \rightarrow \mathbb{R}$$

L.

$$I(f+g) = If + Ig$$

$$C \circ I(f) = C \cdot If$$

12.

1/2 3/4 2/2 0

$$g \in \mathbb{C}. \quad \underline{g = g^+ - g^-}$$

$$\underline{g = g^+ - g^-}$$

$$g = A - B, \quad A, B \in \mathbb{C}^+$$

(20K 3/2 2/2 2/2 2/2 2/2)

u ~ ~

$$Ig = IA - IB \quad \rightsquigarrow$$

2F u.

$$g = A - B = C - D$$

$$A + D = C + B$$

$$I(A + D) = I(C + B)$$

$$IA + ID = IC + IB$$

$$IA - IB = IC - ID$$

2 + 2 ~ Ig 12

well-defined.

$$g = A - B$$

$$-g = B - A$$

$$Ig = -I(-g)$$

$$\forall C \in \mathbb{R}. \quad Ig = C \cdot I(-g)$$

an f to ~

$$f = A - B$$

$$g = D - C$$

$$f + g = A + D - (B + C)$$

$$I(f + g) = I(A + D) - I(B + C)$$

$$= IA + ID - IB - IC$$

$$= If + Ig$$

Haar measure の存在,

$$\cancel{I: L^1}$$

$$\cancel{I: G}$$

$$I: C_0^+(G) \rightarrow \mathbb{R}.$$

$$I(f+g) = I f + I g.$$

$$\forall C \geq 0. \quad I(Cf) = C I f.$$

$$\forall s \in G, \quad I(L_s f) = I f$$

$$L_s f(x) = f(s^{-1}x). \quad \text{つまり } f \text{ は } G \text{ 上の関数.}$$

$$\text{有 } n. \quad s_i \in G. \quad c_i > 0.$$

$$f \leq \sum c_i s_i g.$$

$$f(x) \leq \sum c_i g(s_i^{-1}x).$$

$$(f: g) = \inf \left\{ \sum c_i \right\}.$$

$$\textcircled{1} \quad (L(s)f: g) = (f: g).$$

$$C \geq 0. \quad (Cf: g) = C(f: g).$$

$$\textcircled{3} \quad (f+f': g) \leq (f: g) + (f': g).$$

$$\textcircled{4} \quad (f: g) \leq (f: h)(h: g).$$

$$f \neq 0. \quad f \geq 0.$$

$$(f: g) \geq \frac{\inf f}{\sup f} \Rightarrow \underline{(f: g)} > 0$$

$$\exists \Omega \text{ open s.t. } \inf_{\Omega} f > 0. \quad \text{...}$$

$$(f: g) < \infty.$$

$$f \leq M \Rightarrow f \leq \frac{M}{\inf f} \inf f \leq M \int g.$$

$$\forall x. \quad \inf f > 0. \quad \text{...}$$

$$f \in L_1 \Rightarrow (f: g) \text{ finite.}$$

$$\boxed{f, f_0 \in L^+, \quad 0 \leq \frac{1}{(f_0: g)} \leq \frac{(f: g)}{(f_0: g)} \leq (f: f_0) < \infty.}$$

$$f_0 \notin \mathcal{O} \quad f_0 \in \mathcal{L}.$$

$$A_f(\delta) = \frac{(f, g)}{(f_0, g)}.$$

$$A_f(g) \in J_f = [\sim \sim].$$

$$J = \prod_{f \in \mathcal{L}} J_f$$

$$A_f \delta \in (A_f(\delta))$$

$$F_v = \{ (A_f(\delta)) \mid \text{supp} \delta \subset V, \delta \neq 0 \}$$

$$\{F_v\}_{v \in \mathcal{M}(G)} \text{ is a family of sets}$$

$$\bigcap_v F_v \neq \emptyset.$$

$$\bar{E}_v = F_v \quad I \in \bigcap_v F_v \Rightarrow \dots$$

$$I \in \mathcal{O} \Rightarrow \mathcal{O} \cap \bar{E}_v \neq \emptyset.$$

$$A_{f+g}(\delta) \leq A_f(\delta) + A_g(\delta).$$

$$|I_{f+g} - A_{f+g}(\delta)| < \epsilon.$$

$$|I_f - A_f(\delta)| < \epsilon.$$

$$\begin{aligned} I_{f+g} &< A_{f+g}(\delta) + \epsilon \\ &\leq A_f(\delta) + A_g(\delta) + \epsilon \\ &\leq I_f + I_g + 3\epsilon \end{aligned}$$

$$I_{f+g} \leq I_f + I_g + 3\epsilon.$$

$$I_{f+g} \leq I_f + I_g + 3\epsilon.$$

$$|I_{cf} - A_{cf}(\delta)|$$

$$= |I_{cf} - A_{cf}(\delta)|$$

$$\begin{aligned} |I_{cf} - cI_f| &\leq |I_{cf} - A_{cf}(\delta)| \\ &\quad + |cI_f - A_{cf}(\delta)| \\ &\leq \epsilon + c\epsilon = (1+c)\epsilon \\ &\ll \epsilon, \quad I_{cf} = cI_f. \end{aligned}$$

$$|I_{L_f} - A_{L_f}|$$

$$|I_{L_f} - I_f|$$

$$\leq |I_{L_f} - A_{L_f}(f)|$$

$$\leq |I_f - A_{L_f}(f)|$$

$$< 2\varepsilon$$

$$I_{L_f} = I_f$$

$$I_{f+h} = I_f + I_h, \text{ etc.}$$

$$f \in L^1, \text{ supp } f \subset \mathbb{R}^n$$

$$h, h' \in L^1$$

$$h+h' \in L^1$$

$$s^1, x \in V \Rightarrow$$

$$|h(x) - h(y)| \leq \varepsilon$$

$$|h'(x) - h'(y)| \leq \varepsilon$$

$$s^1, x \in V$$

$$f \in L^1, s^1, x \in V$$

$$h(x) \leq h(s^1) + \varepsilon$$

$$f(x)h(x) \leq \sum' c_i (h(s^1) + \varepsilon) s_i g$$

$$f(x)h(x) \leq \sum c_i (h(s^1) + \varepsilon) s_i g$$

$$(fh)(s) + (f'h')(s)$$

$$\leq (\sum c_i) (1 + 2\varepsilon)$$

$$(fh)(s) + (f'h')(s) \leq (f'h')(1 + \delta)$$

$$A_{fh}(s) + A_{f'h'}(s) \leq A_f(s) (1 + \delta)$$

$$\therefore I_{fh} + I_{f'h'} \leq I_f$$

$$f \in L_+, f' \in L_+$$

$$\text{supp}(f + f') \subset \mathbb{C},$$

$$C \in \mathbb{R}, \inf f'' > 0.$$

$$(f'' \geq \frac{1}{h} \text{ on } C).$$

$$F = f + f' + \varepsilon f''.$$

$$h \in \mathbb{C} \in \mathbb{R}, \frac{f}{F}, \inf f'' = 0.$$

$$h|_C = \begin{cases} \frac{f}{F} & x \in C \\ 0 & x \notin C \end{cases}$$

$$F, h, h' \in L_+.$$

$$h + h' \leq 1.$$

$$hF = f, h'F = f'.$$

$$I_f + I_{f'} \leq I_F.$$

$$I_F \leq I_{f+f'} + \varepsilon I_{f''}.$$

$$I_f + I_{f''} \leq I_{f+f'}.$$

$$I_{f+f'} = I_f + I_{f'}$$

$$\text{supp}(f + f') = K.$$

$$\mathbb{C} \subset V.$$

$$K \subset V \subset \bar{V} \subset \mathbb{R}^n.$$

$$\bar{V} \in \mathbb{R}^n, F = f + f' + \varepsilon f''.$$

$$h = \begin{cases} \frac{f}{F} & x \in V \\ 0 & x \notin V \end{cases}$$

$$F > 0 \text{ (} x \in V \text{)}.$$

$$(X \setminus K) \cap V \in \mathbb{R}^n \text{ is not well-defined.}$$

$$h \text{ is well-defined.}$$

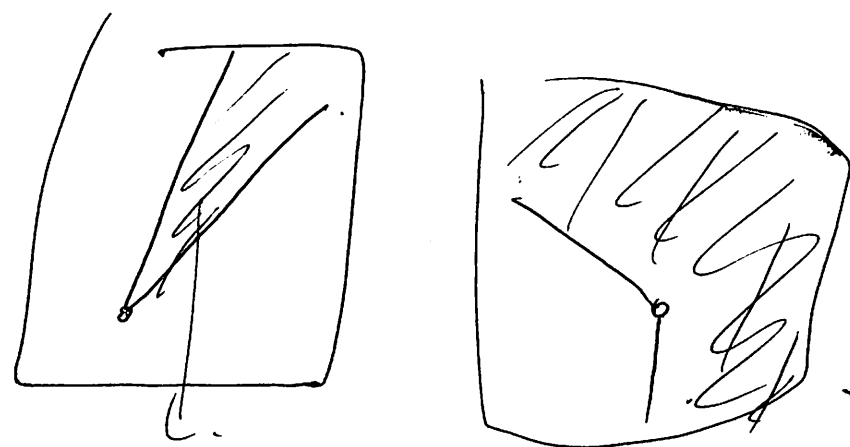
~~注意~~ p. 74.

~~半径の1/2~~

$$Z^n: \mathbb{C} \setminus 0 \rightarrow \mathbb{C} \setminus 0$$

an open map

$z \neq w$



$$\{t e^{i\theta}\}$$

$$\{(r, \theta) \mid 0 < r < \infty, \alpha < \theta - 2 < \varepsilon, r \in \mathbb{C} \setminus 0\}$$

$$Z^n: (r, \theta) \mapsto (r^n, n\theta)$$

$$-n\varepsilon < n\theta - n\alpha < n\varepsilon$$

$$2n\varepsilon < 2\pi\varepsilon$$

$$\varepsilon < \frac{\pi}{n}$$

$$I = (0, \infty) \times (-\varepsilon + 2\pi, 2\pi + \varepsilon)$$

$\varepsilon < \pi$
not
not

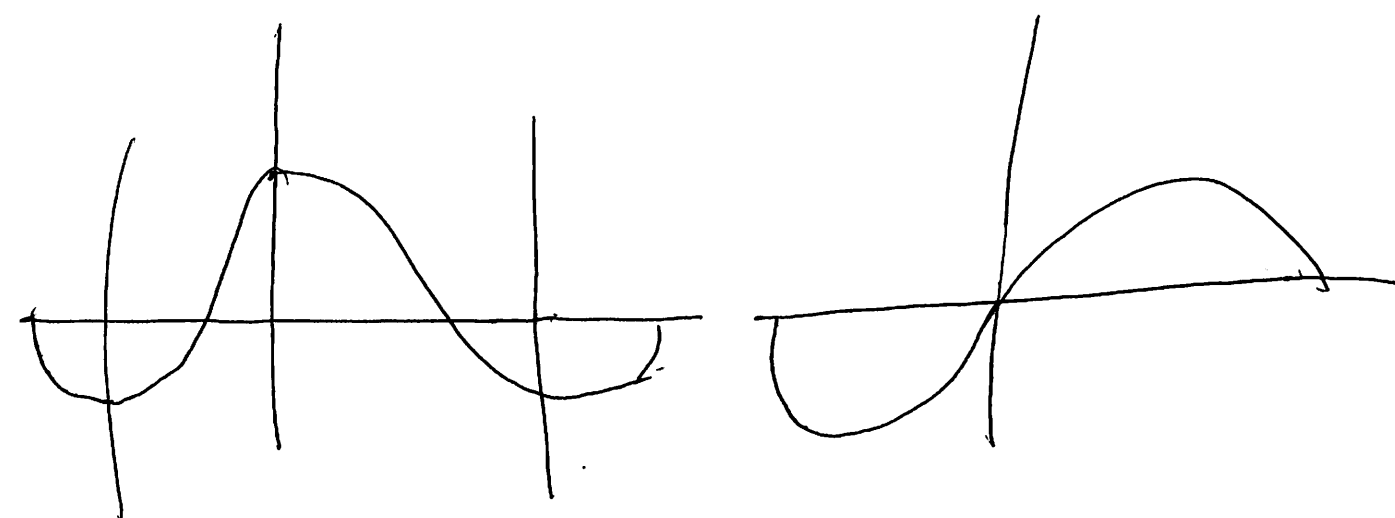
$$E: I \rightarrow \mathbb{C} \setminus \{0\}$$

$$(r, \theta) \mapsto (r e^{i\theta})$$

(ε ∞ $\varepsilon < \pi$
not not)

全射

$$(|z|, \arg z) \mapsto (z)$$



z

$$\{z \mid |z-a| < \varepsilon\}.$$

↓

$$\{z^n \mid |z-a| < \varepsilon\}$$

a^n

$$|z^n - a^n|$$

距離空間 X の Parameter

$\{C_\alpha\}$: open cover.

$$S(x, r), \dots$$

For $n \in \mathbb{N}, \dots$

$$x \in \mathbb{R}^n$$

$$D_{x,n} = \bigcup S(x, 2^{-n})$$

(1). x is the smallest m .
 $x \in C_1$.

(2). $x \notin D_j$ $j < n$.

$$(3). S(x, 3 \cdot 2^{-n}) \subset C_\alpha.$$

$$\{P_{\alpha, n}\}_{\alpha \in A} \subseteq \mathcal{U}_n.$$

$$\mathcal{U}_1 \cup \mathcal{U}_2 \cup \dots = \mathcal{U}_n.$$

$$I_{\alpha} \subseteq I_n \subseteq \mathcal{U}_n.$$

$$(1) \quad x \in \mathcal{U}_n \implies x \in C_2.$$

$$(2) \quad x \notin \bigcup_{j < n} P_{\beta, j} \text{ if } j < n.$$

$$(3) \quad S(x, 3 \cdot 2^{-n}) \subseteq C_2.$$

$$x_1, x_2, \dots, x_n \in \mathcal{U}_n.$$

$$P_{\alpha, n} = \bigcup_{x \in I_{n, \alpha}} S(x, 2^{-n}).$$

$$\text{where } P_{\alpha, n} \text{ are } S. C_2.$$

$$\{P_{\alpha}\}_{\alpha, n}$$

$$x \in \mathcal{U}_n, \quad x \in C_2.$$

$$x \in B_{\beta, j} \text{ for } j < n.$$

$$x \notin P_{\alpha, n}.$$

$$S(x, 2^{-j}) \subseteq P_{\alpha, n}.$$

$$(a) \quad j \geq n+j.$$

$$S(x, 2^{-n-j}) \cap B_{\beta, j} = \emptyset.$$

$$(b) \text{ if } j < n+j. \quad S(x, 2^{-n-j}) \cap B_{\beta, j} \neq \emptyset.$$

$$\underline{j > n.} \quad S(x, 2^{-j}) \not\subseteq P_{\alpha, n}.$$

$$S(x, 2^{-j}) \subseteq P_{\alpha, n}.$$

$$d(x, y) \geq 2^{-j}.$$

$$\hat{i} \geq \hat{j} + 1 \text{ and } n + \hat{j} \geq \hat{j} + 1$$

$$S(x, 2^{-n-\hat{j}}) \cap S(0, 2^{-\hat{i}})$$

$$2^{-\hat{j}} \leq 2^{-n-\hat{j}} + 2^{-\hat{i}} \leq 2^{-\hat{j}-1} + 2^{-\hat{j}-1} = 2^{-\hat{j}}$$

$$-\hat{i} \leq -\hat{j}-1$$

$$2^{-\hat{j}} < 2^{-\hat{j}-1}$$

$$p \in D_{\beta \hat{i}} \quad \perp$$

$$q \in D_{\alpha \hat{i}}$$

$$\beta < \alpha$$

$$d(p, q) > 2^{-n-\hat{j}+1}$$

$$d(p, q) \leq 2^{-n-\hat{j}} + 2^{-n-\hat{j}} = 2^{-n-\hat{j}+1}$$

$$p \in S(\beta, 2^{-\hat{i}}) \subset D_{\beta \hat{i}} \quad \hat{i} < n + \hat{j}$$

$$q \in S(\alpha, 2^{-\hat{i}}) \subset D_{\alpha \hat{i}}$$

$$S(\beta, 3 \cdot 2^{-\hat{i}}) \subset C_{\beta}$$

$$z \notin C(\beta)$$

$$d(z, \beta) \geq 3 \cdot 2^{-\hat{i}}$$

$$\cancel{d(p, q) \leq d(z, \beta)}$$

$$-\hat{i} > -(n + \hat{j})$$

$$3 \cdot 2^{-\hat{i}} \leq d(z, \beta)$$

$$\hat{i} < n + \hat{j} \quad \hat{i} + 1 \leq n + \hat{j} \quad \hat{i} \leq n + \hat{j} - 1$$

$$\leq d(z, \beta) + d(p, \alpha) + d(\alpha, \beta)$$

$$< 2^{-\hat{i}} + 2^{-\hat{i}} + d(p, q)$$

$$2^{-(n+\hat{j}+1)} < 2^{-\hat{i}} < d(p, q)$$

$$2^{-(n+\hat{j}+1)}$$

$$2^{-n-\hat{j}+1} < d(p, q)$$

~~カヤテ~~

~~カ~~

カヤテ、71エルストラ。

$f: \Delta(1) \rightarrow \mathbb{C}$. $z \mapsto z + iz$.

$f(\Delta(1) \setminus \{0\})$ is dense.

Open map th.
 $f \neq \text{const} \Rightarrow f$ is open map.

f is analytic on $\mathbb{C}$.

$$f(0) = 0, \quad |f(z)| < 1.$$

$$\Rightarrow |f(z)| \leq |z|, \quad |f'(0)| \leq 1.$$

$$\exists z \neq 0 \text{ s.t. } |f(z)| = |z| \quad \text{or} \quad |f'(0)| = 1 \text{ then}$$

$$f(z) = e^{i\theta} z.$$

$$\text{Möb} \left\{ \frac{az+b}{cz+d} \mid ad-bc \neq 0 \right\}.$$

$$A = \left\{ az+b \mid a \neq 0 \right\}.$$

$$\sim \text{Möb.} \quad H(z) = \frac{z-i}{z+i}.$$

$$e^{i\theta} \frac{z-a}{\overline{a}z+1} \quad a \in \Delta(1), \quad \theta \in \mathbb{R}.$$

Filter. $\mathcal{F}$ は $\mathcal{P}(A)$ -

$$\mathcal{F} \subset \mathcal{P}(A)$$

$$(i). \emptyset \in \mathcal{F}$$

$$(ii). A \in \mathcal{F}, A \subset B \Rightarrow B \in \mathcal{F}$$

$$(iii). A, B \in \mathcal{F} \Rightarrow A \cap B \in \mathcal{F}$$

$$(3') : A_1, \dots, A_n \in \mathcal{F} \Rightarrow \bigcap_{i=1}^n A_i \in \mathcal{F}$$

$$\emptyset \in \mathcal{F} \Leftrightarrow \mathcal{F} = \mathcal{P}(A)$$

$\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である $\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である

$$\mathcal{F} \text{ は } \mathcal{P}(A) \text{ の } \mathcal{F} \text{ である } A, B \in \mathcal{F} \Rightarrow A \cap B \neq \emptyset$$

$\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である
 $\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である

(ii). $\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である

$$\langle a \rangle = \mathcal{F}$$

$\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である

$$\mathcal{F} \text{ は } \mathcal{P}(A) \text{ の } \mathcal{F} \text{ である}$$

$$\mathcal{F} \text{ は } \mathcal{P}(A) \text{ の } \mathcal{F} \text{ である}$$

$\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である

$\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である

$$\mathcal{F} \subset \mathcal{P}(A)$$

$$\mathcal{F} \text{ は } \mathcal{P}(A) \text{ の } \mathcal{F} \text{ である } \mathcal{F} \cap \mathcal{G} \neq \emptyset$$

$$\mathcal{F} \cap \mathcal{G} = \{ \mathcal{F} \cap \mathcal{G} \mid \mathcal{F} \in \mathcal{F}, \mathcal{G} \in \mathcal{G} \}$$

$\mathcal{F}$ は $\mathcal{P}(A)$ の $\mathcal{F}$ である

极大 $\mathcal{F}_1 \subset \mathcal{A}$ -

$$\square \mathcal{L} \quad \mathcal{F} \subset \mathcal{L} \subset \mathcal{X}$$

$$\Rightarrow \mathcal{F} = \mathcal{L} \text{ or } \mathcal{L} = \mathcal{X}$$

$\mathcal{F}$ 极大

例. $\mathcal{F}$ は 真 $\mathcal{F}_1 \subset \mathcal{A}$ -

① $\mathcal{F}$ は 极大

$$\textcircled{2} \quad \forall A \in 2^X. A \in \mathcal{F} \text{ or } A^c \in \mathcal{F}$$

$$\textcircled{3} \quad A \cap B \in \mathcal{F}$$

$$\textcircled{2} \quad \forall A \in \mathcal{F} \quad A \cap \mathcal{F} \neq \emptyset \Rightarrow A \in \mathcal{F}$$

$$\textcircled{4} \quad A \cup B \in \mathcal{F} \Rightarrow A \in \mathcal{F}, B \in \mathcal{F}$$

$$\textcircled{1} \Rightarrow \textcircled{2} \quad \textcircled{2} \Rightarrow \textcircled{3} \quad \textcircled{3} \Rightarrow \textcircled{4}$$

$$\textcircled{4} \Rightarrow \textcircled{3} \quad \textcircled{3} \Rightarrow \textcircled{1}$$

极大 $\mathcal{F}_1 \subset \mathcal{A}$ - の 性質

极大 $\mathcal{F}_1 \subset \mathcal{A}$ - の 像

filter

$$\lim \mathcal{F} = \mathcal{a}$$

$$\Leftrightarrow \pi(\mathcal{a}) \subset \mathcal{F}$$

Filter の 概念

$\mathcal{C}_A$ 上の 順序関係

$$\mathcal{X} \in \mathcal{C}_A \Leftrightarrow \text{内点}$$

$\mathcal{F}_1 \subset \mathcal{F} \subset \mathcal{C}_A$ 上の

$$\textcircled{1} \quad \mathcal{X} \in \mathcal{C}_A$$

$$\textcircled{2} \quad \mathcal{F} \subset \mathcal{F}_1 \dots \exists \mathcal{G} \quad \mathcal{F} \subset \mathcal{G} \quad \mathcal{L} \neq \mathcal{F}_2$$

$$\textcircled{3} \quad \mathcal{X} \in \mathcal{C}_A \text{ 上 の } \mathcal{F}_1 \subset \mathcal{A} - \mathcal{F} \text{ は } \mathcal{F}_2 \text{ ではない}$$

Tychonov の 定理

$$\sum_n \mathcal{C}_A \Rightarrow \prod \mathcal{X}_n \in \mathcal{C}_A$$

$$p \in D_{\beta, i}, q \in D_{\gamma, i} \quad \beta < \gamma.$$

$$d(p, q) > 2^{-n-j+1}.$$

$$\text{Set } x \in I_{\beta, i} \quad y \in I_{\gamma, i}.$$

$$\text{Set } p \in S(x, 2^{-i}), \\ q \in S(y, 2^{-j}).$$

$$\beta < \gamma \text{ then}$$

$$\text{if } y \in D$$

$$\text{then } y \in C_\gamma \text{ then } y \in C_\beta.$$

$$y \notin C_\beta \text{ then}$$

$$y \notin D_{\beta, i} \text{ then}$$

$$S(x, 3 \cdot 2^{-i}) \subset C_\beta \text{ then}$$

$$d(x, y) \geq 3 \cdot 2^{-i}.$$

$$3 \cdot 2^{-i} \leq d(x, y)$$

$$\leq d(x, p) + d(p, q) + d(q, y)$$

$$< 2 \cdot 2^{-i} + d(p, q).$$

$$2^{-i} < d(p, q) \quad i < n+j-1$$

$$i+1 \leq n+j.$$

$$i \leq n+j-1. \quad -i \geq -n-j+1.$$

$$2^{-n-j+1} \leq 2^{-i} \leq d(p, q).$$

$$I, 2. \quad 2^{-n-j+1} \leq d(p, q)$$

$$41. \quad S(x, 2^{-n-j}) \cap D_{\beta, i} \cap D_{\gamma, i}$$

$$2^{-n-j+1} \leq d(p, q) \leq 2^{-n-j} + 2^{-n-j} \geq 2^{-n-j+1}.$$

$$\text{if } x \text{ is not in } D, \quad p, q = \min \quad \square$$

$$(p, q) = \{x, p < x < q\}.$$

$$(p \rightarrow) = \{x, p < x\}.$$

$$(\leftarrow q) = \{x, x < q\}.$$

$$\text{convex } x, p, q \in V$$

$$\Rightarrow [p \wedge q, p \vee q] \subset V$$

$$\text{convex set, open set } V, \text{ if } V \subset \mathbb{R}^n.$$

$$\text{if } x \in O \Rightarrow \exists V \text{ convex open } x \in V \subset O.$$

LOTS IS $\{ \sigma(\Sigma, \leq) \}$

H, K : separated. $H \cap K = \emptyset$. $H \cap \bar{K} = \emptyset$.

$x \in H$. $\therefore \cancel{x \in K} \exists v. x \in V_x \subset \Sigma \setminus \bar{K}$ (open).

$x \in K$. $x \in V_x \subset \Sigma \setminus H$.

$$V_H = \bigcup_{x \in H} V_x \quad V_K = \bigcup_{x \in K} V_x$$

$$H \subset V_H \subset \Sigma \setminus K$$

$$K \subset V_K \subset \Sigma \setminus H$$

$$V = V_H \cap V_K \subset \Sigma \setminus (H \cup K)$$

$$V = \emptyset \quad \text{or} \quad V \neq \emptyset$$

$V \neq \emptyset$. $p \sim q$

$$\Leftrightarrow [p \wedge q, p \vee q] \subset V$$

Σ is not 11.

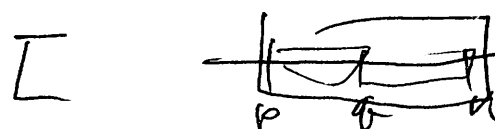
$$p \sim p \text{ ok.}$$

$$p \sim q \Rightarrow q \sim p \text{ ok.}$$

$$p \sim q \wedge q \sim k \Rightarrow p \sim k$$

$k < p < q$. $p < k < q$ or $q < k < p$ or $q < p < k$.

$$p < q < k \dots$$



$0 < k$

$\uparrow \Sigma$. $\sim$ is a complete representation of Σ .

$$x \in H \text{ i.e. } p, q \in V_x \cap T, \quad p < q$$

$$\Rightarrow p < x < q \text{ or } q < x < p$$

$$x < p < q \text{ i.e. } p \in V_x \text{ and } q \in V_x$$

$$p \in V_x \Rightarrow [p, q] \subset V_x$$

$$p, q \in V_x \text{ i.e. } y \in [p, q]$$

$$p < x \text{ i.e. } y < x < p. \quad y \in [p, q] \subset V_x$$

$$\text{or } y \perp, \text{ i.e. } x < y, \Rightarrow x < p < q < y$$

$$p \in [p, q] \subset V_x. \quad [p, q] \subset V_x$$

$$[p, q] \subset V_x$$

$$1. [p, q] \subset V_x \cap V_y \subset V$$

$$p \sim q \text{ or } y \perp, \text{ i.e. } p < x$$

$$\text{or } p < x < q$$

$$\therefore |V_x \cap T| \leq 2$$

$$\left(|V_x \cap T| \geq 3 \text{ i.e. } p < q < r \text{ or } p < x < q < x < r \text{ or } q < x < p < x < r \right)$$

$$W_x = \begin{cases} V_x & \text{if } x \in H \cup K \\ V_x \cap (P_i \rightarrow) & \text{if } x < \alpha \\ V_x \cap (\leftarrow P) & \text{if } x < \beta \\ V_x \cap (P_i \vdash) & \text{if } x < \gamma \end{cases}$$

$$W_H = \bigcup_{x \in H} W_x$$

$$W_x \cap T = \emptyset$$

$$W_\alpha = \bigcup_{x < \alpha} W_x$$

$$H \subset W_H, \quad K \subset W_K - \text{obvious}$$

$$W_H \cap W_\alpha = \emptyset \text{ if } \alpha \in H$$

$$W_x \cap W_y \neq \emptyset \text{ if } x < y \Rightarrow x < y < z$$

$$y \in V, y \sim p, p \in T \text{ if } y < z$$

$$I \subset V \subset X \setminus (H \cup K)$$

$$x < p < z, \quad p < \alpha, \quad p < \beta, \quad p < \gamma$$

$$p \in W_\alpha, \quad p \in W_\beta, \quad p \in W_\gamma$$

$$p \in W_\alpha, \quad p \in W_\beta, \quad p \in W_\gamma$$

$$\rightarrow W_H \cap W_N = \emptyset$$

$$W_x = \begin{cases} V_x & \text{if } x \notin H \cup K \\ V_x \cap (p_i \rightarrow) & \text{if } x < p_i \\ V_x \cap (\leftarrow p) & \text{if } x < p \\ V_x \cap (p, q) & \text{if } p < x < q \end{cases}$$

$$W_H = \bigcup_{x \in H} W_x$$

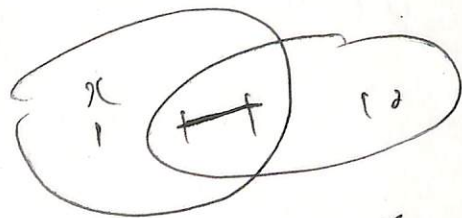
$$W_K = \bigcup_{x \in K} W_x$$

$$W_x \cap T = \emptyset$$

$$H \subset W_H, K \subset W_K \text{ - obvious}$$

$$W_H \cap W_K = \emptyset \text{ if } \exists x \in H, y \in K$$

$$W_x \cap W_y \neq \emptyset \text{ if } x < y \text{ or } y < x$$



$$I \subset [p, q], p < q$$

$$I \subset V \subset X \setminus (H \cup K)$$

$$x < y$$

$$I \neq \emptyset$$

$$x < p < q$$

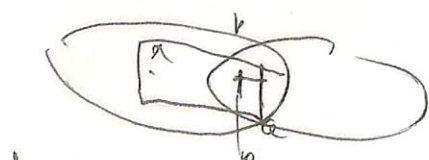
$$p < x$$

$$p < x < q$$

$$p < q < x$$

$$p < q < x$$

$$p < q < x$$



$$p < q < x$$

$$p < q < x$$

$$p < q < x$$

$$\rightarrow W_H \cap W_K = \emptyset$$